

Design Report
Bike Handle Analysis



Prepared by:
Isabella Fuierer
Jennifer Gjonaj
Michaela Nanna
Aiden Samperi

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Department of Mechanical Engineering
School of Engineering and Computing
Fairfield University

Introduction

For our project, we chose to analyze a bicycle handlebar subjected to static structural loading. The handlebar model consists of a horizontal curved tube with a vertical stem that attaches to the bicycle frame. This configuration was selected because handlebars experience a combination of forces during real-world use, including vertical loads from the rider's body weight, lateral forces from steering inputs, and dynamic vibrations from varying road surfaces. Analyzing the structural integrity of the handlebar under these loading conditions is essential to ensure rider safety, material durability, and optimal performance.

In our setup, the base of the stem was treated as a fixed support to simulate the clamped connection to the bicycle frame. Loads were applied at the ends of the handlebars to represent the rider's hands pushing and pulling during normal riding. A fine mesh was applied throughout the model, with local mesh refinement around curved sections and the junction between the handlebar and stem, where stress concentrations were expected.

We selected several materials for evaluation, including structural steel, titanium alloy, and carbon fiber composite, which are commonly used in modern bicycle designs due to their favorable strength-to-weight ratios. Before conducting the simulation, the handlebar appeared structurally sound for normal riding conditions, but detailed finite element analysis allowed us to predict deformation behavior, stress distributions, and safety factors, providing critical insight into the mechanical performance of different material options.

Analysis

I. Assumptions:

- The handlebar and stem connection are perfectly rigid at the mounting point (fixed support).
- The rider is aggressively cycling, resulting in a force of 280 N on the handlebar.
- The applied loads represent static forces only, neglecting dynamic effects such as vibration or impact.
- Forces from the rider's hands are evenly distributed across both ends of the handlebar.
- Material properties, such as Young's modulus and yield strength are homogeneous and isotropic for each material tested.
- Temperature effects and thermal expansion are neglected.

- The handlebar geometry has no manufacturing defects or imperfections.

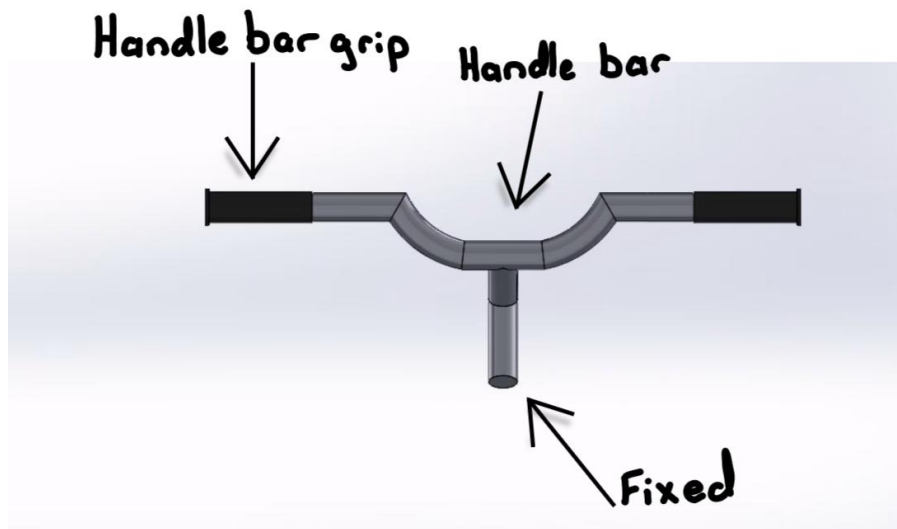


Fig. 1. Shows a bike handlebar model in SolidWorks and includes the rubber handlebar grip. Part is an assembly to differentiate the handlebar grips from the handle.

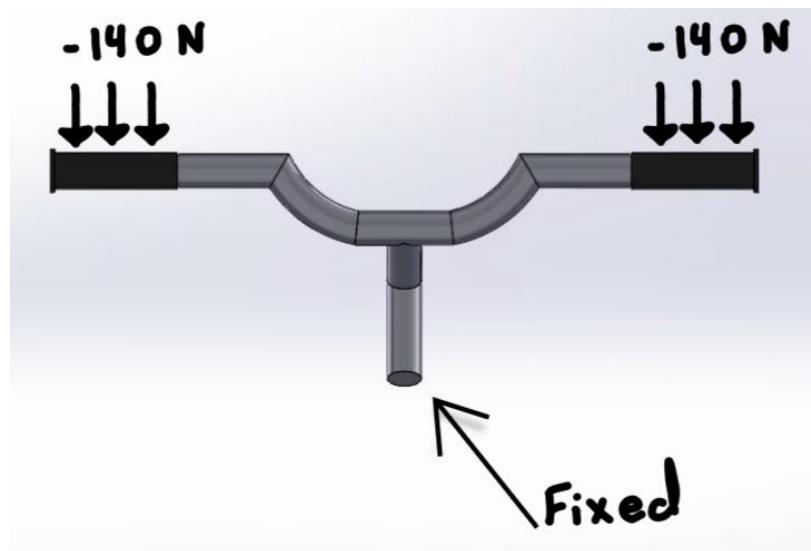


Fig. 2. Shows the Free body diagram and forces that were used in ANSYS to perform calculations to get the stress and deformation.

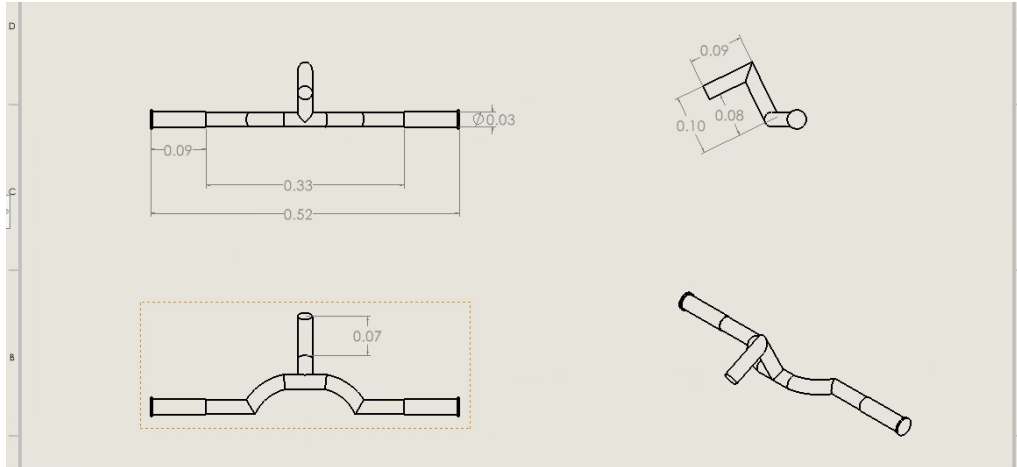


Fig. 3. Shows the engineering drawing with the bike handle and handlebar grips. The units are in centimeters.

II. Methodology

When setting up the bicycle handlebar for analysis, we aimed to observe how different material combinations would affect stress, deformation, and safety factor under typical rider loading. For simplification, we modeled only the handlebar and stem, eliminating smaller components such as grips and fasteners. The bottom of the vertical stem was selected as the fixed support, simulating the connection point to the bicycle frame.

To simulate the load of a rider gripping the handlebars, we applied downward forces at each end of the handlebar. These loads were distributed as nodal forces across two rectangular regions, one on each grip area, representing the pressure from the rider's hands. The regions were placed symmetrically from the center to reflect typical riding posture.

Fine mesh was applied to the entire model to improve accuracy, with additional refinement in the curved elbow sections of the handlebar where prior simulations showed higher stress concentrations. This refinement helped improve result accuracy in critical regions of bending.

We repeated this simulation for four material combinations: Carbon Fiber–Carbon Fiber, Carbon Fiber–Structural Steel, Titanium–Carbon Fiber, and Titanium–Structural Steel. Material properties were assigned using standardized values for elastic modulus,

Poisson's ratio, density, and yield strengths, as outlined in the Appendix. After solving, we collected data on total deformation, principal stress, and safety factor for comparison.

III. Improvements

Based on the analysis results, several enhancements can be considered to optimize the handlebar design. If budget permits, utilizing advanced materials such as carbon fiber or titanium could lead to a lighter and thinner handlebar, potentially improving performance. However, it's crucial to conduct thorough testing to ensure that these materials maintain the desired ride quality, particularly regarding the handlebar's flex and vibration dampening properties. Additionally, modifying the handlebar's geometry, especially in areas with sharp curves or transitions, may help reduce stress concentrations. Redesigning these sections to have smoother transitions could distribute stress more evenly and enhance durability. Lastly, evaluating and potentially reinforcing the connection between the handlebar and the stem could address areas where higher stress concentrations were observed, ensuring a more reliable overall structure.

Results

In the first analysis, the entire handlebar is made from carbon fiber. Carbon fiber is chosen because it offers an excellent balance of high strength and low weight, making it ideal for applications like handlebars where minimizing weight improves performance without sacrificing durability

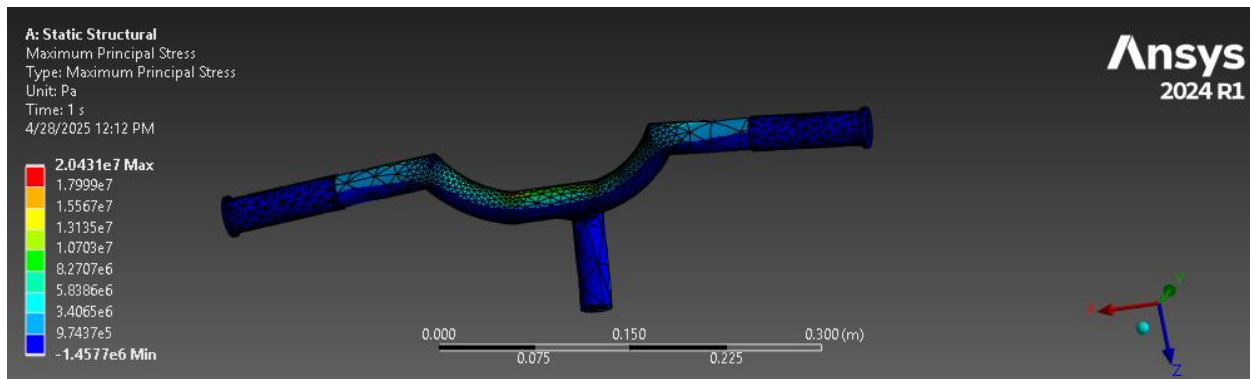


Fig. 4. Maximum Principal Stress for Carbon Fiber Handlebar

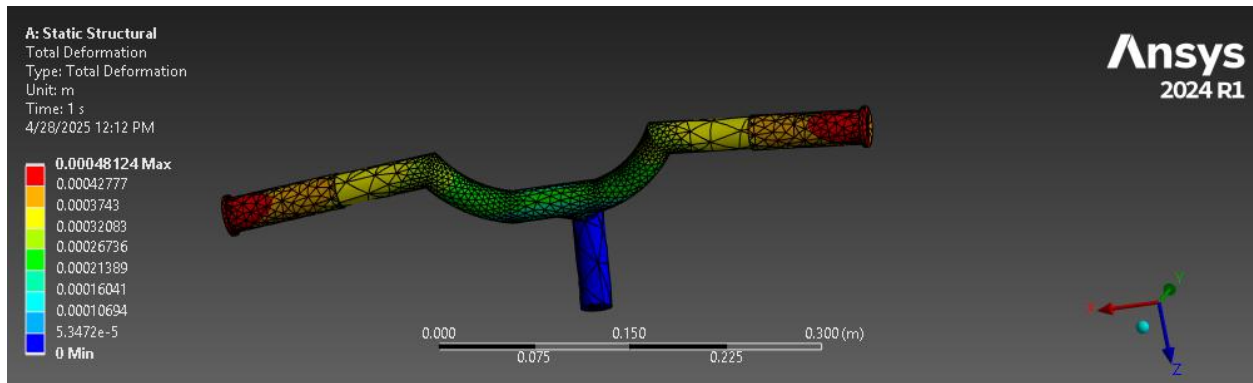


Fig. 5. Total Deformation for Carbon Fiber Handlebar

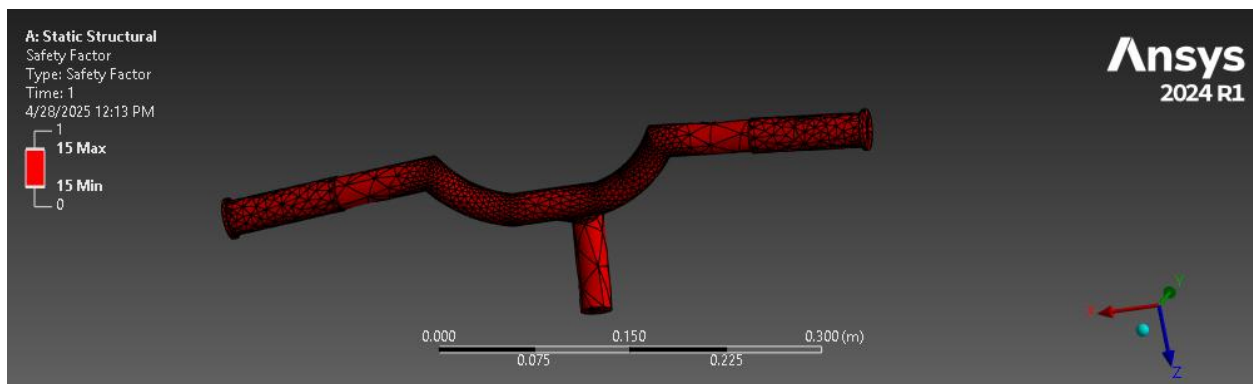


Fig. 6. Safety Factor for Carbon Fiber Handlebar

In this analysis, the handlebar uses a combination of carbon fiber and structural steel. Carbon fiber provides lightweight strength where weight savings are critical, while structural steel is used to add durability and toughness in areas that experience higher stresses.

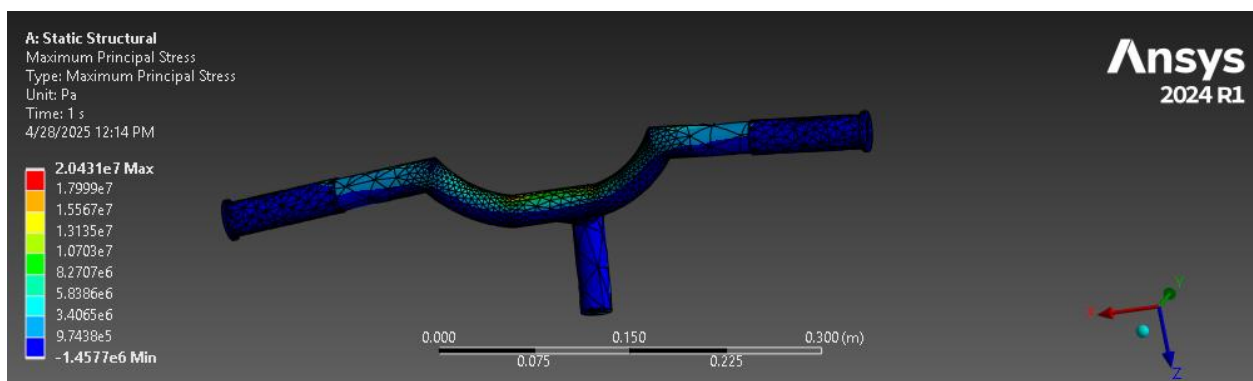


Fig. 7. Maximum Principal Stress for Carbon Fiber and Structural Steel Handlebar

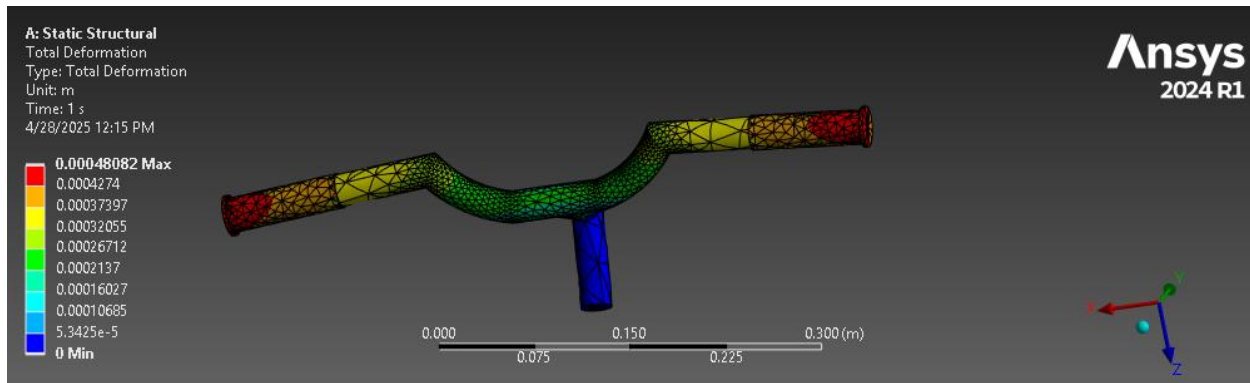


Fig. 8. Total Deformation for Carbon Fiber and Structural Steel Handlebar

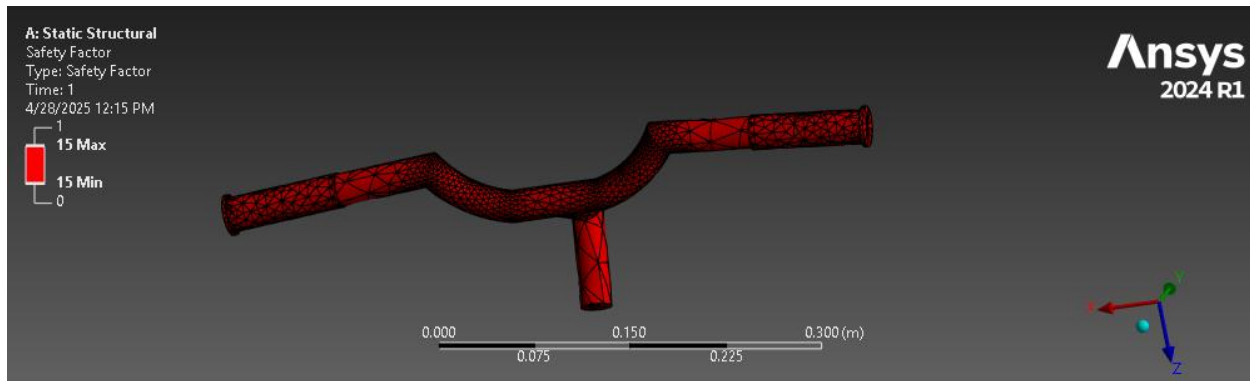


Fig. 9. Safety Factor for Carbon Fiber and Structural Steel Handlebar

In this analysis, the handlebar combines titanium and carbon fiber. Carbon fiber keeps the handlebar lightweight and stiff for better performance, while titanium adds exceptional strength, fatigue resistance, and corrosion resistance.

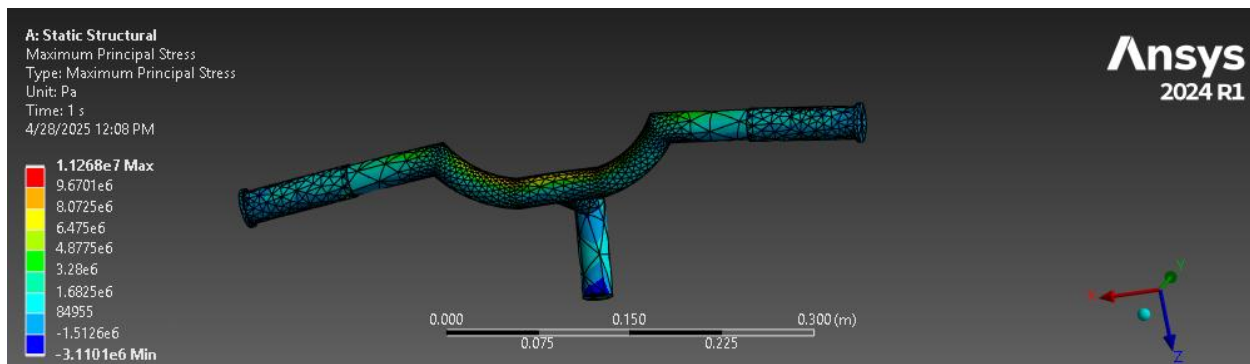


Fig. 10. Maximum Principal Stress for Titanium and Carbon Fiber Handlebar

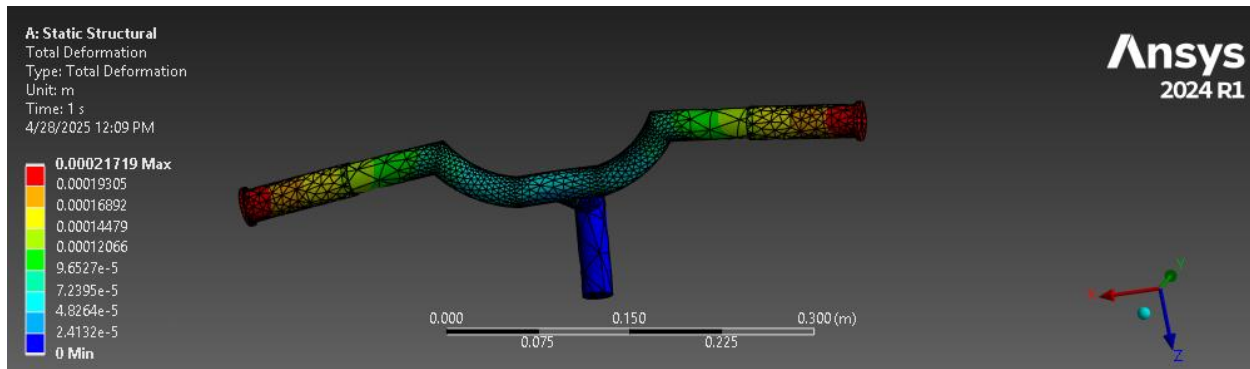


Fig. 11. Total Deformation for Titanium and Carbon Fiber Handlebar

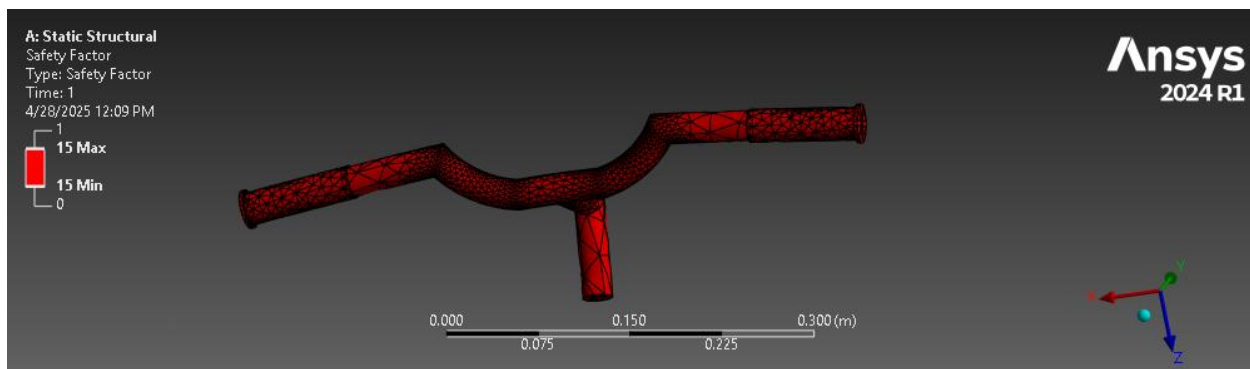


Fig. 12. Safety Factor for Titanium and Carbon Fiber Handlebar

In this analysis, the handlebar combines titanium and structural steel. Titanium provides excellent strength, light weight, and corrosion resistance, while structural steel offers additional toughness and cost-effectiveness.

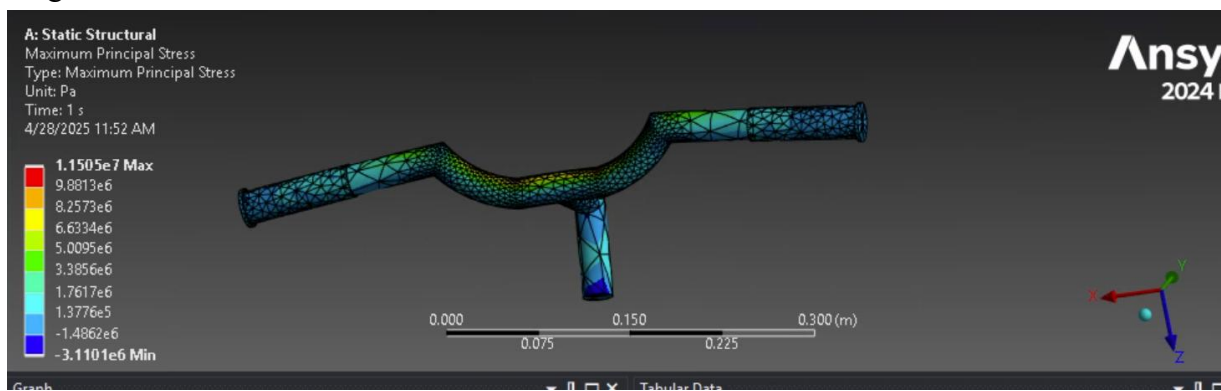


Fig. 13. Maximum Principal Stress for Titanium and Structural Steel Handlebar

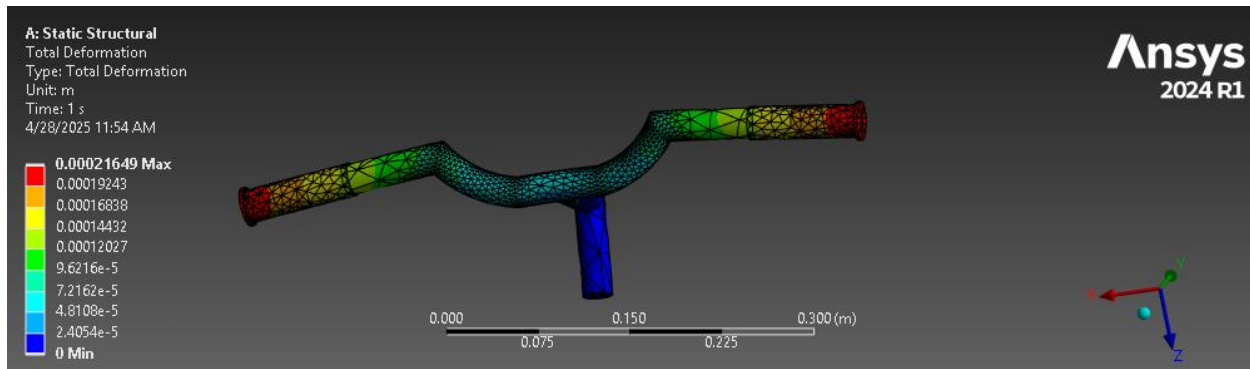


Fig. 14. Total Deformation for Titanium and Structural Steel Handlebar

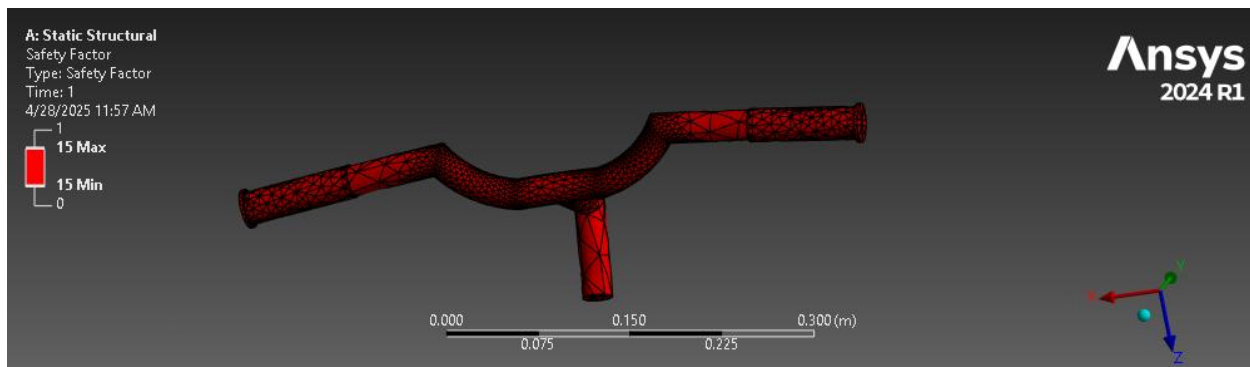


Fig. 15. Safety Factor for Titanium and Structural Steel Handlebar

Table 1. Summary of ANSYS simulation results showing maximum deformation, minimum and maximum principal stress, and safety factors for different material combinations used in the handlebar analysis.

Material Combination	Max Deformation (m)	Min Stress (Pa)	Max Stress (Pa)	Min Safety Factor	Max Safety Factor
Carbon Fiber-Carbon Fiber	0.000216	- 1457100	20431000	15	15
Carbon Fiber – Structural Steel	0.000481	- 1457100	20431000	15	15
Titanium-Carbon Fiber	0.000481	- 1457100	20431000	15	15

Titanium- Structural Steel	0.000217	-3110100	15050000	15	15
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Discussion

The finite element analysis conducted on the bicycle handlebar across four material configurations—carbon fiber-carbon fiber, carbon fiber-structural steel, titanium-carbon fiber, and titanium-structural steel— provides valuable insights into how material selection affects mechanical performance under typical loads.

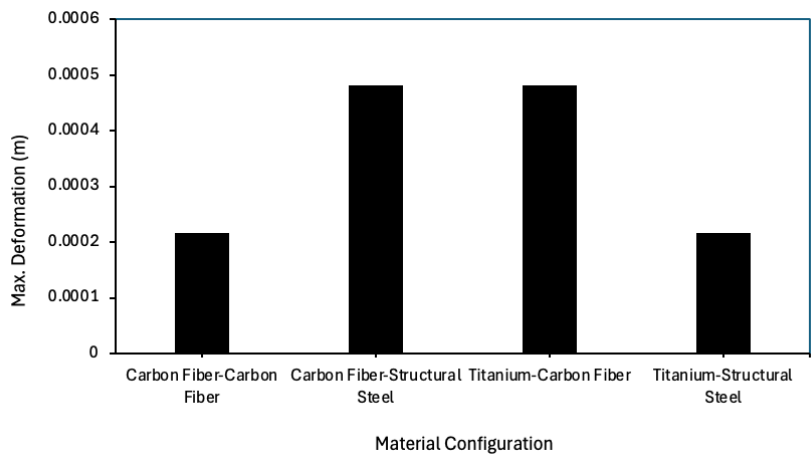


Fig. 16. Bar graph comparing maximum deformation values for each material configuration.

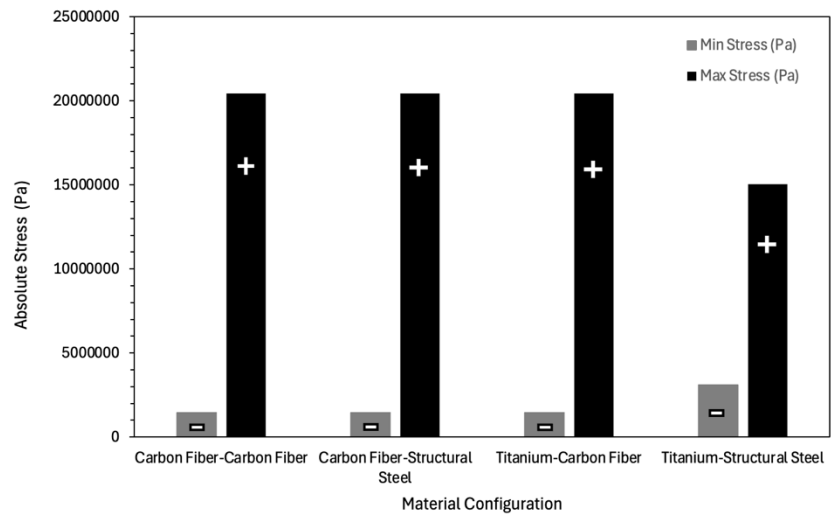


Fig. 17. Bar graph comparing absolute stress values for each material configuration, with negative signs indicating a negative stress value, and a positive sign indicating a positive stress value.

The carbon fiber-carbon fiber configuration exhibited the lowest total deformation, 0.000216 m, while maintaining a maximum stress of 20.43 MPa and a safety factor of 15. This indicates that carbon fiber provides great stiffness and strength for the loads applied, which validates its widespread use in bicycle components that are performance focused. Although carbon fiber is lightweight and strong, it can be both brittle and costly, which are important considerations when applying to the real-world. The titanium-structural steel configuration also showed minimal deformation, 0.000217 m, which is comparable to the carbon-fiber only setup. However, this configuration experienced a higher maximum stress of 15.05 MPa. Despite this, the safety factor remained at 15 across all materials, which suggests that all configurations are within acceptable limits for static loading conditions. Titanium's excellent fatigue resistance and corrosion resistance make it a strong candidate, although it also comes at a high manufacturing cost.

The hybrid configurations, carbon fiber-steel and titanium-carbon fiber, had higher deformations of 0.000481 m, likely due to the mismatch in elastic moduli between the combined materials. This mismatch can result in uneven stress distribution and potential stress concentrations at material interfaces. These findings reinforced the importance of uniform material properties or careful bonding strategies in designs using composite materials.

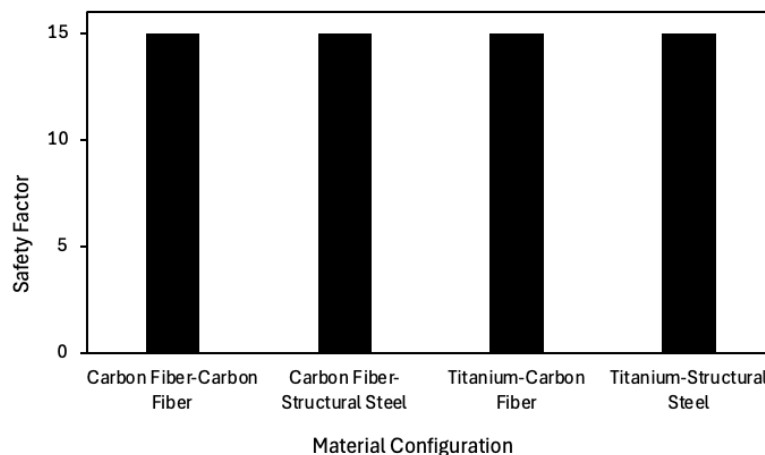


Fig. 18. Bar graph comparing safety factor values for each material configuration.

The consistent safety factor of 15 across all configurations suggests that the current handlebar geometry is over-designed for the applied load. While this ensures safety, it can present opportunities for optimizations, such as reducing material usage or modifying geometry, without compromising structural integrity. Further studies with dynamic loading or fatigue analysis could offer deeper insights into long-term performance.

Conclusion

Through detailed static structural analysis using ANSYS, we evaluated the performance of four material combinations in a bicycle handlebar model. All tested materials produced safety factors of 15, which indicates that each configuration exceeds the strength requirements under the assumed 280 N load. Among them, the carbon fiber-carbon fiber configuration showcased the best balance of low deformation and stress, thus confirming its effectiveness for high-performance applications. Despite similar safety factor results, considerations such as cost, manufacturability, fatigue resistance, and comfort guide choice in material. Titanium and structural steel offer higher toughness and resistance, while carbon fiber delivers superior stiffness to weight ratios at a good price. While all designs are structurally sound for static use, the carbon fiber-carbon fiber setup is most suitable for applications prioritizing lightweight performance, and the titanium-structural steel configuration may be more practical for cost-effective durability. Future analysis under dynamic or fatigue loading should be conducted to fully validate these conclusions for real-world applications

Appendix

Table A.1 Properties of Carbon Fiber

Property Name	Value	Units	Value Type
Elastic modulus	7.00E+10	Pa	Constant
Poisson’s Ratio	0.1	-	Constant
Shear Modulus	3.18E+10	Pa	Constant
Mass Density	1600	kg m^3	Constant
Tensile Yield Strength	6.00E+08	Pa	Constant
Compressive Yield Strength	3.24E+08	Pa	Constant

Table A.2 Properties of Titanium

Property Name	Value	Units	Value Type
Elastic modulus	9.60E+10	Pa	Constant
Poisson's Ratio	0.36	-	Constant
Shear Modulus	3.53E+10	Pa	Constant
Mass Density	4620	kg m ³	Constant
Tensile Yield Strength	1.07E+09	Pa	Constant
Compressive Yield Strength	9.30E+08	Pa	Constant

Table A.2 Properties of Structural Steel

Property Name	Value	Units	Value Type
Elastic modulus	2E+11	Pa	Constant
Poisson's Ratio	0.30	-	Constant
Shear Modulus	7.69E+10	Pa	Constant
Mass Density	7850	kg m ³	Constant
Tensile Yield Strength	2.5E+08	Pa	Constant
Compressive Yield Strength	2.5E+08	Pa	Constant